Conditional random fields improve local ancestry model predictions

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Summary

Background: Local ancestry inference (LAI) identifies the ancestry of each segment of an individual genome and has several uses in medical and population genetics. Many tools have been developed for LAI, including ones using random forests[1] and hidden Markov models. Our group previously proposed a model based on the SegNet architecture, which has the advantage that a trained model can be applied to different input datasets (e.g. multiple cohorts with varied array designs) without the need for re-training because of the SegNet's positional invariance. However this approach fails to correctly predict genomic ancestries in regions in which haplotypes are not associated with ancestry.

Methods: Here, we propose a conditional random field (CRF) to post-process the SegNet's predictions, using sequential information, which the SegNet cannot. The CRF is trained on the SegNet's predictions for admixed samples simulated from 372 real haplotypes.

Results: The CRF boosts the classification of genetic variants on haplotypes made by 10 generations of simulated admixture from 85.6% to 95.1%.

Conclusions: We successfully formulate LAI as a segmentation problem and find that adding a CRF to SegNet brings the overall accuracy near that of the current gold standard RFMix (97.5% accuracy over same test set).

Methods

A linear-chain CRF (Fig. 1) is employed where the predicted probabilities of each class are used as features for each position. We attempted adding recombination probabilities between positions as features, but these did not boost our accuracy.

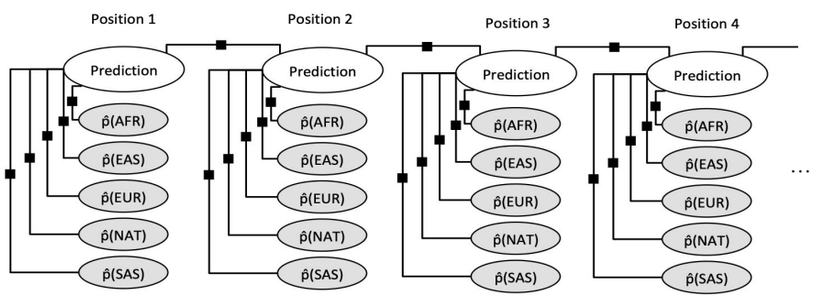

Figure 1: Pictorial representation of linear-chain CRF. White ovals represent predictions and shaded ovals represent class probabilities predicted by SegNet.

Dataset

We use 2,522 chr20 haplotypes from unrelated persons in the 1000 Genomes, Simons Genome Diversity, and Human Genome Diversity Projects. Ground truth labels are assigned with an unsupervised clustering technique (ADMIXTURE[3]), and individuals' haplotypes are put randomly into train/dev/test sets.

	African AFR (n=734)	East Asian EAS (n=988)	European EUR (n=308)	Native American NAT (n=150)	South Asian SAS (n=342)
Train	578	778	236	102	272
Test	52	80	24	16	22

Table 1: Train/dev/test set size by population. Note $n_{test} = n_{dev}/2$ for all ancestry groups.

Results

Model output and ground truth for a sample individual (Figure 2); confusion matrixes for SegNet with and without CRF post-processing (Tables 2/3). Additional experiments can be found on our project GitHub[4].

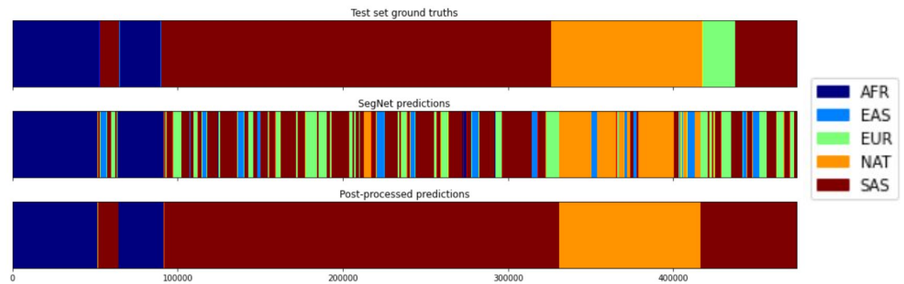

Figure 2: True local ancestry (top), above model predictions for a sample haplotype (bottom/middle are with/without CRF). Axis label is variant index along chr20.

Tables 2/3: Fraction of base pairs of true ancestries (rows) assigned to each group (column) by SegNet (left) or SegNet with CRF post-processing (right). Values are for 100 haplotypes simulated by 10 generations of pan-continental admixture. Diagonal entries are ancestry-specific sensitivity; net accuracy is 85.6% and 95.1%.

SegNet ⇨ / Truth ⇩

	AFR	EAS	EUR	NAT	SAS
AFR	0.955477	0.007248	0.016773	0.000799	0.019703
EAS	0.003523	0.932756	0.005511	0.005105	0.053105
EUR	0.046745	0.046744	0.648671	0.010746	0.247094
NAT	0.002929	0.173253	0.041182	0.721892	0.060744
SAS	0.040143	0.186164	0.176702	0.012062	0.584929

SegNet +CRF ⇦

	AFR	EAS	EUR	NAT	SAS
AFR	0.969364	0.007987	0.013238	0.001559	0.007851
EAS	0.004679	0.973322	0.003991	0.002791	0.015217
EUR	0.010653	0.020174	0.851846	0.005978	0.111349
NAT	0.003359	0.024859	0.029521	0.931060	0.011202
SAS	0.009899	0.032236	0.008745	0.006249	0.942871

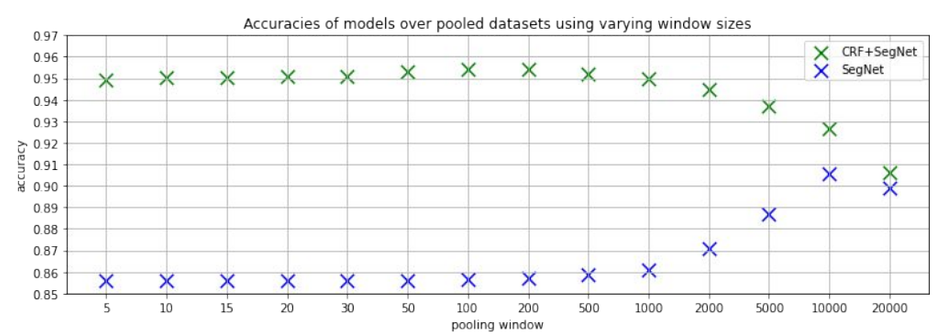

Figure 3: Model accuracies after training over simplified datasets where windows of varying sizes were collapsed into single positions for which the SegNet's predicted class probabilities were averaged.

References

1: Maples *et al.*, *AJHG* '13.
2: Badrinarayanan *et. al.*, *ArXiv* '16.
3: Alexander *et al.*, *Bioinformatics* '11.
4: https://github.com/maguirre1/deepLAI/